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Paralelismo



3º Grado en Ingeniería Informática



Facultad de Informática de Barcelona (Fib)
Universidad Politécnica de Catalunya



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Qualificació 4,00 sobre 4,00 (100%)

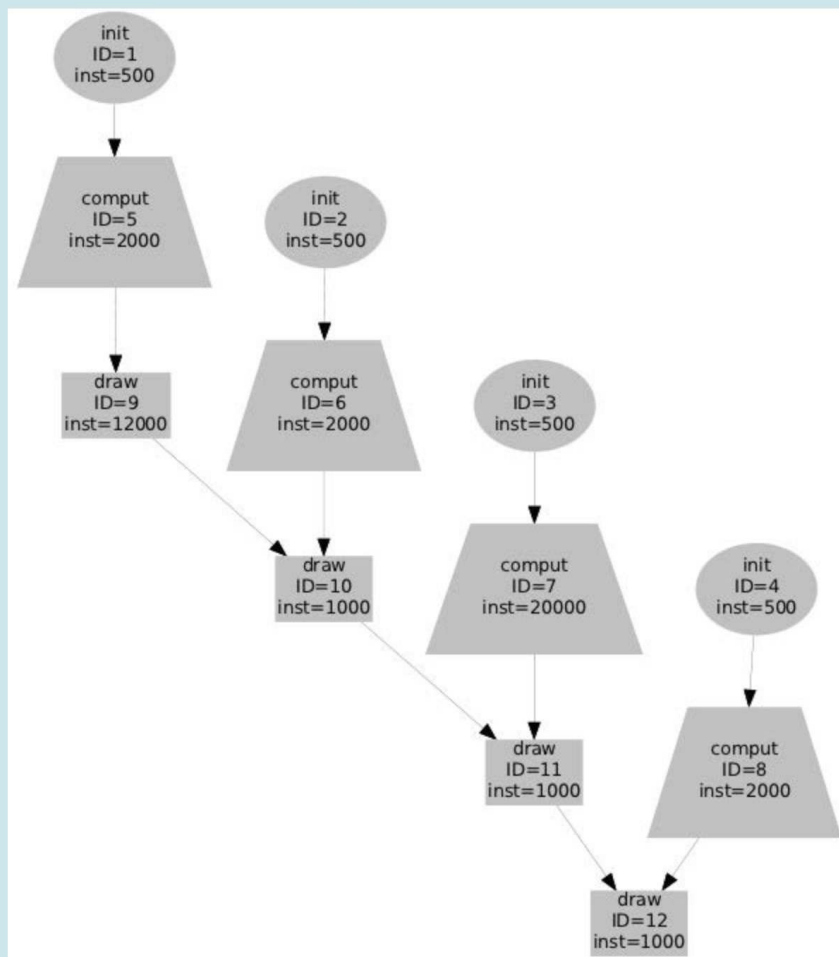
Pregunta 1

Completa

Puntuació 1,00
sobre 1,00

Marca la
pregunta

Given the following task dependence graph (TDG). Each node is labeled with a name and also includes an identifier *ID* and cost *inst* in terms of the number of instructions.



The TDG can be represented equivalently with the following table:

WUOLAH

ID=12
inst=1000

The TDG can also be expressed textually with the following table:

name	ID	inst	sucesor ID task
init	1	500	5
init	2	500	6
init	3	500	7
init	4	500	8
compute	5	2000	9
compute	6	2000	10
compute	7	20000	11
compute	8	2000	12
draw	9	12000	10
draw	10	1000	11
draw	11	1000	12
draw	12	1000	No successor

Observe that task *comput* with *ID*=7 takes 10 times more that the other *comput* tasks and task *draw* with *ID*=9 takes 12 times more than other *draw* tasks.

Assume that the tasks in the TDG are executed on 4 processors with the following task assignment: each processor executes a sequence *init-compute-draw* (for example the sequence {2, 6, 10}). Which is the speed-up that is obtained?

Tríeu-ne una:

- ☒ 1.91
- ☐ 2.45
- ☐ 4

Pregunta 2

Completa

Puntuació 1,00
sobre 1,00 Marca la
pregunta

Assuming that we are able to better balance the work among processors, which means that each node 1-4 weights 500, each node 5-8 weights 6500, and each node 9-12 weights 3750. Which is the speed-up that would be achieved with 4 processors, assuming the same task assignment as before?

Trieu-ne una:

- ☐ 4
- ☒ 1.95
- ☐ 4.77

Pregunta 3

Completa

Puntuació 1,00
sobre 1,00 Marca la
pregunta

Assume a sequential application computing the sum of two vectors of size $N=1024$ elements. Which should be the problem size and task granularity when parallelized with $P=4$ processors and strong scaling:

Trieu-ne una:

- ☒ 1024 and 256, respectively.
- ☐ 1024 and 1024, respectively.
- ☐ 4096 and 1024, respectively.

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Pregunta 4

Completa

Puntuació 1,00
sobre 1,00

[Marca la pregunta](#)

Which should be the problem size and task granularity when parallelized with $P=4$ processors and weak scaling:

Trieu-ne una:

- ☐ 1024 and 256, respectively.
- ☐ 4096 and 256, respectively.
- ☒ 4096 and 1024, respectively.

WUOLAH